Notes: Equiangular Tight Frames via Geometric Invariant Theory

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January 2026

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Chapter 1

Preliminaries

1.1 Motivation:

Let V be a K vector space where K is algebraically closed.

Definition 1. *Let $Fr_n(V) := \{(v_i)_{i=1}^n \in V^n \mid v_i \neq \lambda v_j,$ for $i \neq j$, $span(\{v_i\}) = V\}$ be called set of frames.*

We define some actions of different groups on $Fr_n(V) \times Fr_n(V^*)$ to get a notion of equivalent frames:

- $GL(V)$ acts on the left as : $M((v_i), (\phi_i)) = ((Mv_i), (v \mapsto \phi_i(M^{-1}v)))$
- $(K^\times)^n \rtimes S_n$ acts on the right as (scaling and permuting frames):
  - $((v_i), (\phi_i))\lambda_i = ((\lambda_i v_i), (v \mapsto \phi_i(\lambda_i^{-1}v)))$
  - $((v_i), (\phi_i))\sigma = ((v_{\sigma(i)})(\phi_{\sigma(i)}))$

Definition 2. *Now we need to look at the subspace of $GL(V)\backslash Fr_n(V) \times Fr_n(V^*)/(K^\times)^n \rtimes S_n$ called $\boldsymbol{ETB_n(V)}$ which satisfies the following:*

1. $\phi_i(v_i) = 1$
2. $\phi_i(v_j)\phi_j(v_i) = \frac{n-d}{d(n-1)}$ *for* $i \neq j$
3. $v_j = \frac{d}{n}\sum_{i=1}^{n} \phi_i(v_j)v_i$

<u>Now the **BIG QUESTIONS** are:</u>

- what is $ETB_n(V)$? Variety?
- when is dimension of $ETB_n(V)$ zero? In that case, what is the number of points $\#ETB_n(V)$?

Special Case: Equiangular Tight Frames

$K = \mathbb{C}$ complex numbers, $\phi_i = v_i^\dagger \implies \phi_i(v) = \langle v_i, v \rangle = v_i^\dagger v$ is called **ETF**.

1. unit norm: $\langle v_i, v_i \rangle = 1$
2. equiangular: $| \langle v_i, v_j \rangle |^2 = \dfrac{n-d}{d(n-1)}$
3. tight: $v = \frac{d}{n} \sum\limits_{i=1}^{n} \langle v_i, v \rangle v_i$

Chapter 2

Polarization and Restitution:

2.1 Invariant functions, quotients, and finite generation

2.1.1 Invariant functions.

Let k be an algebraically closed field and let G be an algebraic group acting on an affine variety $X = \mathrm{Specm}A$, where A is a finitely generated k-algebra. The action of G on X is equivalently given by a rational action of G on A by k-algebra automorphisms. We define the ring of *invariant functions* by

$$A^G := \{f \in A \mid g \cdot f = f \text{ for all } g \in G\}.$$

Remark 1 (Geometric meaning). *An invariant function is constant along G-orbits. Thus A^G measures exactly the functions on X that cannot distinguish points inside a single orbit. The philosophy of GIT is that the quotient space should have coordinate ring A^G.*

2.1.2 Spec vs. Specm.

There are two closely related spaces associated to a ring A^G:

$$\mathrm{Spec}A^G \quad \text{and} \quad \mathrm{Specm}A^G.$$

The maximal spectrum $\mathrm{Specm}A^G$ corresponds to closed points, while $\mathrm{Spec}A^G$ includes also generic points encoding specialization relations.

Remark 2. *In algebraic geometry, $SpecA^G$ is the correct object for forming quotients. Even when one is interested in closed orbits (points), non-closed orbits play a crucial role, and their closures are naturally encoded in $SpecA^G$, not in $SpecmA^G$.*

Example 1. *Let $G = \mathbb{Z}/2$ act on $A = k[x, y]$ by*

$$\sigma(x) = y, \qquad \sigma(y) = x.$$

Then the invariant ring is

$$A^G = k[x + y,\, xy].$$

The morphism

$$\pi : \mathbb{A}^2 \to \mathbb{A}^2, \qquad (x, y) \mapsto (x + y, xy)$$

is G-invariant and induces an isomorphism

$$Speck[x + y, xy] \cong SpecA^G.$$

Geometrically, π identifies the points (x, y) and (y, x), and separates distinct orbits except when (x, y) and (y, x) lie in the same orbit closure. In particular, invariant functions separate closed *orbits but may fail to separate non-closed ones.*

Remark 3 (Key phenomenon)**.** *The fibers of π are precisely orbit closures, not individual orbits. This is the fundamental reason the GIT quotient is a* categorical *quotient rather than a geometric one.*

2.1.3 Affine schemes and schemes.

An *affine scheme* is a space of the form Spec$A$ for a ring A. A *scheme* is obtained by gluing affine schemes along open subsets. Thus schemes are locally affine objects.

Remark 4. *For an affine scheme $X = SpecA$, one has a canonical identification*

$$A \cong \mathcal{O}_X(X).$$

This property characterizes affine schemes among all schemes.

2.1.4 Quotients and finite generation.

Given an affine G-variety $X = \mathrm{Spec}A$, a natural candidate for the quotient is

$$X//G := \mathrm{Spec}A^G.$$

This construction only makes sense if A^G is finitely generated as a k-algebra.

Theorem 3 (Hilbert, classical form)**.** *If G is a reductive algebraic group acting rationally on a finitely generated k-algebra A, then A^G is finitely generated.*

Remark 5 (Why reductivity matters)**.** *Reductive groups have enough invariant linear functionals to control the growth of invariants. For non-reductive groups, invariant rings can be pathologically large.*

2.1.5 Hilbert's 14th problem and Nagata's counterexample.

Hilbert asked whether A^G is finitely generated for *every* linear action of an algebraic group. This is false.

Theorem 4 (Nagata)**.** *There exists a linear action of a unipotent algebraic group G on a polynomial ring A such that the invariant ring A^G is not finitely generated.*

Remark 6 (Conceptual takeaway)**.** *Finite generation is not a formal property of group actions; it is a deep structural fact tied to reductivity. GIT works precisely because it restricts attention to reductive group actions, where quotients exist as affine schemes of finite type.*